

Hasna Hena Jui

hasna.hena.jui@g.bracu.ac.bd

Dhaka, Bangladesh

LinkedIn

Phone: +8801339659260

Research Focus

VLSI Design, Compute-in-Memory (CiM), Neuromorphic Computing, Memory-Centric Architectures, Emerging Memory Devices.

Education

BRAC University, Dhaka, Bangladesh

(Expected Summer 2026)

Bachelor of Science in Electrical and Electronic Engineering

Specialization: Electronics

CGPA: **3.61/4.0**

Publications

- **H. H. Jui**, M. T. A. Tonmoy, M. R. Kabbo, A. A. Mahi, N. M. Himel, S. Sultana, T. Titirsha, and T. Rahman, "Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks," *IEEE International Symposium on VLSI (ISVLSI)*, 2026. (Accepted)

On-Going Work

- **SRAM-Based Compute-in-Memory MAC Accelerator.** (Current)
Designing a scalable SRAM compute-in-memory (CIM) accelerator for multiply-accumulate (MAC) operations in neural network workloads. The research focuses on SRAM compute-cell architecture, peripheral circuit design, sensing schemes, and energy-efficient in-memory computation to reduce data movement and improve AI inference performance.
- **RRAM Compute-in-Memory Modeling and Crossbar Analysis.** (Current) Developing Verilog-A compact models for resistive random-access memory (RRAM) devices and investigating 1T1R-based compute-in-memory architectures.

Research Experience

- **Compute-in-Memory** (2026–Present)
Variation-Aware SRAM Compute-in-Memory Architecture for Binary Neural Networks Designed and simulated a 45 nm 6T SRAM-based compute-in-memory (CIM) architecture in Cadence Virtuoso for in-memory Boolean computation. Implemented AND, OR, NAND, NOR, XOR, and XNOR operations and performed Monte Carlo simulations to quantify the impact of process variations on delay and logic reliability. Developed a Python/PyTorch-based evaluation framework to investigate the relationship between circuit-level timing variations and binary neural network inference accuracy.
SRAM-Based CIM MAC Accelerator Developing a scalable SRAM compute-in-memory accelerator for multiply-accumulate (MAC) operations in neural network workloads. Current work focuses on compute-cell array organization, peripheral circuit design, accumulation architectures, and the evaluation of energy, latency, and reliability trade-offs.
- **Emerging Memory and Device Modeling** (2026–Present)
Verilog-A Based RRAM Compact Modeling Investigating resistive random-access memory (RRAM) devices and developing compact behavioral models using Verilog-A. Studying resistance-switching dynamics, HRS/LRS characteristics, and SET/RESET mechanisms to support future compute-in-memory and neuromorphic computing architectures.
- **Final Year Design Project / Capstone Project** (2026–Present)
Autonomous Agricultural Rover for Precision Farming Developing an autonomous rover for grid-based soil sampling and zone-specific fertilization in precision agriculture applications. The project integrates NPK sensing, navigation, motion planning, and simulation-based performance evaluation using robotic simulation environments, with the goal of enabling data-driven fertilizer management.

Course and Other Projects

- **Design and Simulation of a Two-Stage CMOS Operational Amplifier.** (2026)
Designed and simulated a two-stage CMOS operational amplifier in 45 nm GPDK technology using Cadence

Virtuoso. Performed transistor-level sizing, DC operating-point analysis, AC frequency response characterization, stability assessment, and transient simulations to evaluate gain, bandwidth, phase margin, slew rate, and power consumption.

- **BRACU Diganta: CanSat Communication and Flight Systems.** (2024–Present)
Contributed to the communication subsystem of BRACU Diganta, a global finalist in the CanSat Competition 2025 (Virginia, USA). Designed ground-station antennas, integrated XBee telemetry links, conducted link-budget analysis, and developed aerodynamic and auto-gyro descent models. Currently serving as Project Manager, leading the team and overseeing communication subsystem development for the Teknofest 2026 Model Satellite Competition.
- **CNN-Based Floating Waste Classification on Water Surfaces.** (2026)
Developed and optimized CNN-based image classification models for automated detection of six categories of floating aquatic waste. Conducted data preprocessing, augmentation, training, and performance evaluation to support autonomous environmental monitoring systems.
- **Design and Development of a Satellite Training Kit.** (2025)
Developed a LoRa based ground station for an educational satellite platform for real-time telemetry reception, decoding, and visualization through an Arduino Nano. Implemented communication protocols and packet handling for satellite-to-ground link demonstrations.

Awards and Achievements

- **Merit Based Scholarship** – Awarded a 25% tuition scholarship for academic excellence during Summer 2022 and Fall 2022.
- **Dean's List Award** – Recognized on the Dean's List for outstanding academic performance in three semesters.
- **CanSat Competition 2025 Global Finalist** – American Astronautical Society, Virginia, USA. Selected among the global finalist teams to participate in the international CanSat Competition.
- **Outstanding FYAT Mentor** – Summer 2025. Recognized for exceptional mentorship, academic guidance, and contributions to supporting first-year students' academic and professional development.
- **International Conference Publications** – IEEE ISVLSI 2026. Co-authored research papers accepted at the IEEE Computer Society Annual Symposium on VLSI (ISVLSI 2026).

Leadership and Community Involvement

- **Student Mentor, First Year Advising Team (FYAT)** (Summer 2024–Spring 2026)
Office of Academic Advising, BRAC University. Provided academic mentoring and peer support to incoming undergraduate students.
- **Project Manager, BRACU Diganta** (2024–Present)
Leading multidisciplinary teams in the design, development, testing, and competition preparation of CanSat systems, coordinating subsystem integration and project execution.
- **Vice President, BRAC University Electrical and Electronic Club** (2026–Present)
Leading technical initiatives, workshops, and student development programs while promoting research engagement and professional growth within the EEE community.

References

- **Twisha Titirsha, PhD**
Assistant Professor, Department of Electrical and Electronic Engineering.
BSRM School Of Engineering, BRAC University.
Email: twisha.titirsha@bracu.ac.bd